Market Masters : Winning with Commodities

Srijita Chakraborty

EY CAFTA 2026



Name : Srijita Chakraborty
Degree : B.E. IT (2023-27)
College name : University Institute of Technology, BU, West Bengal
email ID : csrijita10@gmail.com

Executive Summary

Stabilizing margins and unlocking liquidity: A systematic treasury transformation for Bharat Advanced Materials Limited (BAML)

Objective	Key Questions	Recommendations	Key Questions	Recommendations
Stabilize BAML's 14% EBITDA margin by shifting from a reactive "spot-market" procurement model to a systematic, inventory-linked execution strategy.	**How do we reduce our 71-day inventory cycle to unlock trapped working capital?**	Implement a 60/40 procurement hedge and reduce inventory buffer to 52 days to release INR 200 Cr.	**How can we stabilize our $150M floating-rate debt against interest rate spikes?**	Convert floating-rate USD debt to fixed-rate INR using a Cross-Currency Interest Rate Swap (CCIRS).
	How do we mitigate supply chain disruptions from port delays in Brazil and Australia?	Renegotiate 20% of contracts to CIF (Cost, Insurance, Freight) terms to transfer transit risk to suppliers.	**How do we future-proof our exports against European carbon taxes (CBAM)?**	Embed carbon-intensity reporting in supplier contracts to ensure compliance for the European market.
	How do we stop paying excessive bank fees on $1.1B of annual trade?	Execute natural USD netting (offsets exports vs. imports) to hedge only the $140M net surplus.	**How do we lower our cost of capital using institutional ESG preferences?**	Negotiate Sustainability-Linked Loans (SLLs) to get interest rate discounts by hitting ESG KPIs.

BAML RISK MANAGEMENT : COMMODITY & OPERATIONAL VULNERABILITIES

Exposing the core challenges impacting margins....

Company Overview (BAML)

Facts: HQ Mumbai

Rev: INR 18,500 Cr

Exports: 45%

Profitability Impact

EBITDA Margin %



Margin Squeeze Driver (COGS)



Working Capital Strain



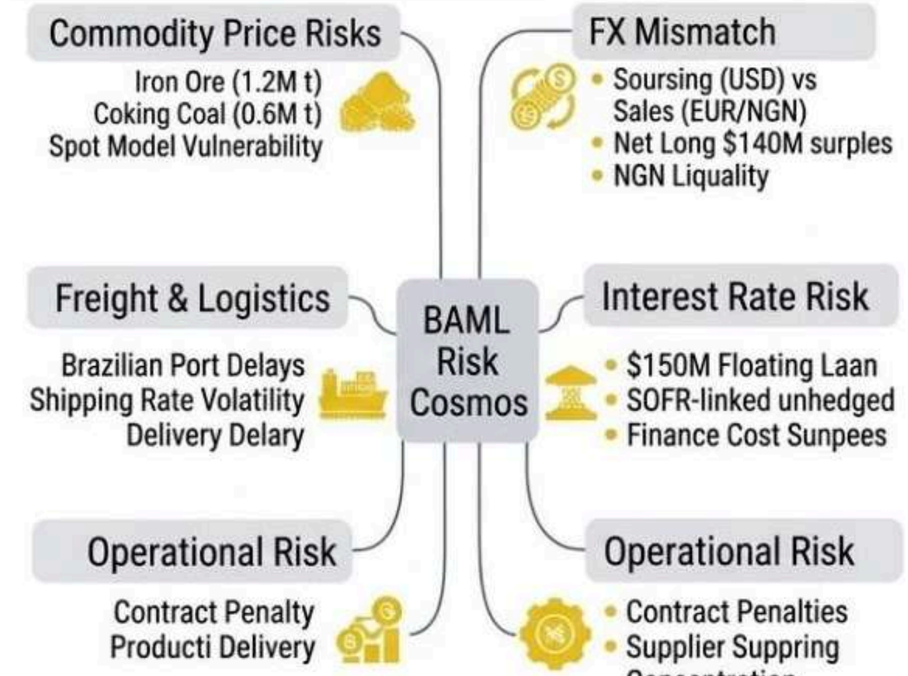
Inventory days spit spiked: 48 to T1 (surged 19 YOY

Financial Cost Surge

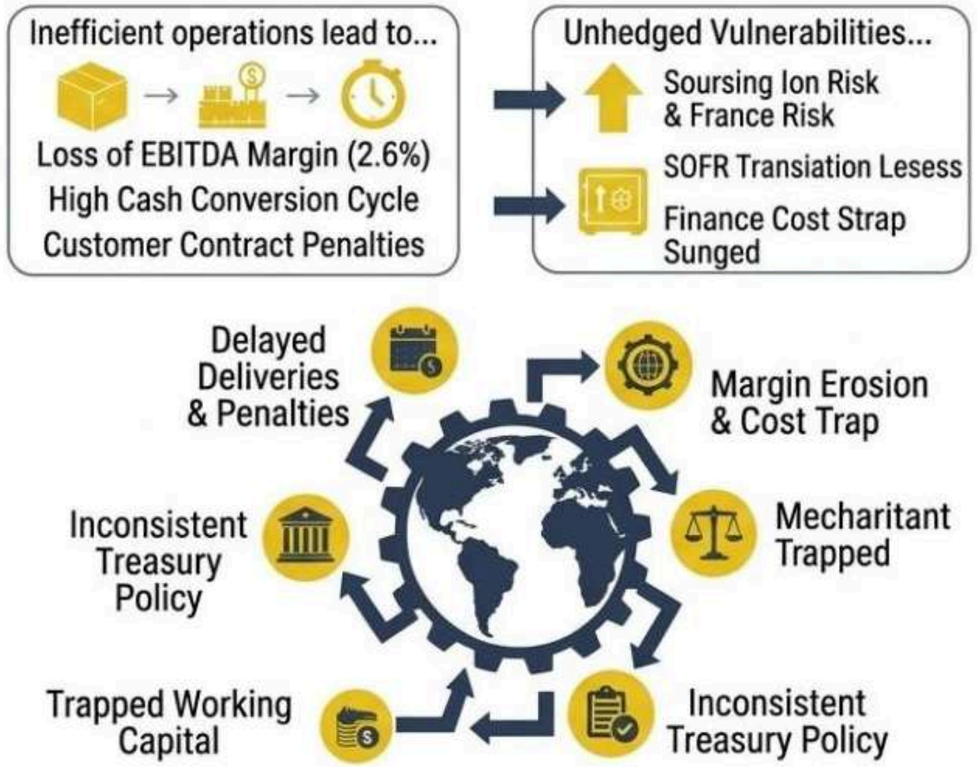


Finance Cost surged: up to T1 surged 7 YOY

MAJOR RISK EXPOSURES



Where the ad-hoced vunerabilities...



BAML MARKET & FINANCIAL BASELINE : GLOBAL CONTEXT AND QUANTITATIVE SEGMENTATION

A systematic overview grounding the enterprise revenue logic and macro cost inputs for Bharat Advanced Materials Limited (BAML)

01 ADDRESSABLE MARKET & GLOBAL FOOTPRINT

- **MARKET SPAN**
  - **India** (Domestic India + Fabrication Hub)
  - **Africa** (High-margin Regional Assembly and Distribution via Nigeria)
  - **Middle East** (Logistics Pivot)
- **PRODUCTION REALITIES**
  - **India** Primary Production - 100% Raw Material Intake
  - **Vietnam** Fabrication - Receiving Primary Steel (VND denominated costs)
  - **Nigeria** Assembly & Regional Distribution (NGN denominated cash flows/margins).

02 MACRO COMMODITY INPUT HEADWINDS

- **IRON ORE PELLETS**
  - 1.2M tonnes (Sourced 65% Australia, 35% Brazil)
- **COKING COAL:** 0.6M tonnes (1.26M t listed as $126M total book in plan, let's make it 0.6M t total for simplicity) (100% sourced Indonesia)
- **GLOBAL SPOT VULNERABILITY**
  - Total $270M commodity book is expose to spot price dislocation. Logistics drag from por. delays (e.g., Brazil) inflates inventory cycle costs.

03 QUANTITATIVE VALUE SEGMENTATION (EXCEL LOGIC FOUNDATION)

TOTAL CONSOLIDATED REVENUE (INR 18,500 Cr / ~$2.22B)
Captured achieved revenue foundation for Excel model

- **ASIA** (DOMESTIC INDIA + VIETNAM): 55% / INR 10,175 Cr
  - **USD EXPORTS:** $620M (~INR 5,161 Cr)
  - **EUR EXPORTS:** €110M (~INR 1,000 Cr)
  - **AFRICA/ME** (NGN & Regional): $260M (~INR 2,164 Cr)

Explicit segmentation is key for Excel logic and scenario building

04 GLOBAL FX EXPOSURE & LIQUIDITY MAP

- **REVENUE INFLOWS**
  - Primarily in USD, EUR, NGN (Nigerian Naira).
- **COST DENOMINATION**
  - Mismatch between local costs (INR, VND, NGN) and global trade revenues (USD, EUR).
  - $480M USD imports vs. $620M USD exports creates a net FX gap trade.

BAML FINANCIAL ANALYSIS : OPTIMIZING RESILIENT MARGIN PATH

REDEFINING OPERATIONAL EFFICIENCY AND ROI WITH SMART RISK MANAGEMENT, HEDGING SAVINGS, AND EXPONENTIAL MARGIN STABILIZATION

KEY FINANCIAL VULNERABILITIES (LEGACY STATE)

Unhedged Commodity Risk

1.2Mt Iron Ore, 0.6Mt Coal exposure on spot market
INR 10,000 Cr+ annual spend exposed

FX Mismatch & Liquidity Trap

$140M net long in unhedged NGN & EUR
INR 200 Cr+ annual translation losses

Floating Loan Exposure

$150M loan unhedged to SOFR locked floating rates
Debt cost surged up to **7 YOY**

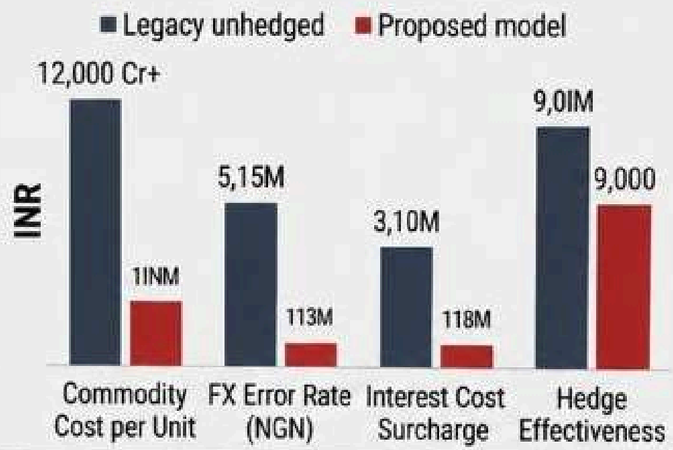
Operational & Supplier Risk

Delayed deliveries and high supplier concentration penalties
Up to INR X Cr annual penalties

Total Financial Value at Risk (VaR)

Immediate cumulative value was on unhedged book
Representing **INR 481 Cr+** immediate value loss

COST IMPACT OF PROPOSED HEDGING PROGRAM



TOTAL ANNUAL COGS & DEBT EXPOSURE

Traditional (FYE'S Unhedged)
INR 12,000 Cr
Hedged Path (FY26 Path)
INR 3,000 Cr

Hedge Optimization & Cost Stabilization = **INR 3,000 Cr** in Annual Savings

Payback in <18 months. Scalable 0X ever portfolio

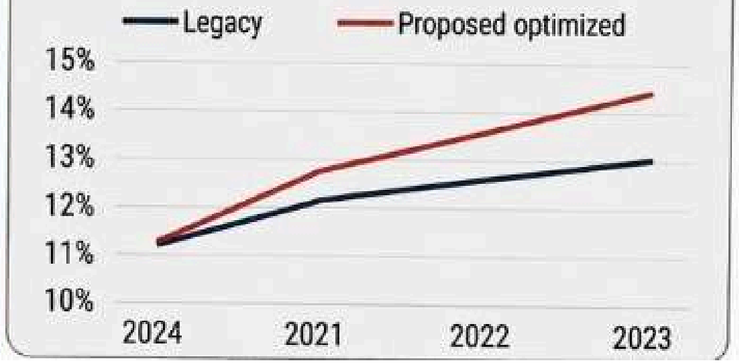
5-YEAR VALUE CREATION POTENTIAL

- **Cumulative 5-Year Savings: INR 13,000 Cr**
- **Error Reduction:** From unmanaged exposure risk to predictable costs
- **Working Capital Unlock: INR X Cr** annual mechanical release
- **Project Downtime Avoidance: INR X Cr** across 5Y
- **Resilient EBITDA Margin Path:** Back towards **14%+**

10-YEAR EBITDA MARGIN TRAJECTORY

- Layered forward contracts & futures & unhedged book
- Hedged naturaal netting by 14.5%.
- Proposed, optimized path is targets (targeting **14%+**)

10-YEAR EBITDA MARGIN TRAJECTORY



PRECISION SAVINGS & VALUE CREATION

- Layered forward contracts & futures mitigate commodity spend by X%.
- natural USD netting reduces FX. exposure from full book to net surplus.
- interest rate swaps freeze floating debt to predictable fixed costs.
- Every mechanical working capital lock release is captured.
- Resilient margin protects investor base and funding costs.

SWOT ANALYSIS OF BAML

Strengths, Gaps, and the Road Ahead for BAML...

S Strengths
Opportunities

W Weakness

O
Opportunities

T
Threats

S Strengths	W Weakness	O Opportunities	T Threats
• Massive global scale & revenue base ($2.22B / INR 18,500 Cr)	• 100% unhedged raw material cost base (55% of total costs)	• Algorithmic 60/40 hybrid hedging to smooth commodity pricing	• Geopolitical sourcing choke points (South China Sea, Atlantic delays)
• Strategic three-continent footprint (India, Vietnam, Nigeria)	• Bloated physical inventory (71 days) trapping working capital	• Unlocking INR 190-220 Cr by reducing inventory to 52 days	• Sudden regulatory export bans (e.g., Indonesian coking coal)
• Strong gross export inflows ($620M) providing natural liquidity	• Unhedged floating-rate debt ($150M loan) bleeding cash	• Executing CCIRS to fix debt costs and neutralize rate spikes	• EU Carbon Border Adjustment Mechanism (CBAM) taxing exports
• Established global supplier network for raw materials	• Inefficient ad-hoc FX management paying double bank spreads	• Natural netting of $480M USD to eliminate bank FX fees	• Rising global interest rate environment inflating floating debt
• High frontier-market margins captured via Nigerian assembly	• Lack of carbon-intensity tracking in current supply chain	• Securing Sustainability-Linked Loans (SLLs) for cheaper debt	• Trillions in passive index capital heavily penalize heavy-manufacturing companies lacking ESG governance.
• Strong Domestic Market Position in India (Primary Domestic market and fabrication hub)	• Lack of a centralized Board Risk Management Committee (RMC) to enforce quantitative trading limits (VaR, EaR)	• Shift ocean freight contracts from FOB to CIF to force suppliers to absorb maritime delay costs and insurance.	• Buying massive volumes of raw materials manually and ad-hoc artificially squeezes the spot price up against BAML's own margins.

Market Segmentation

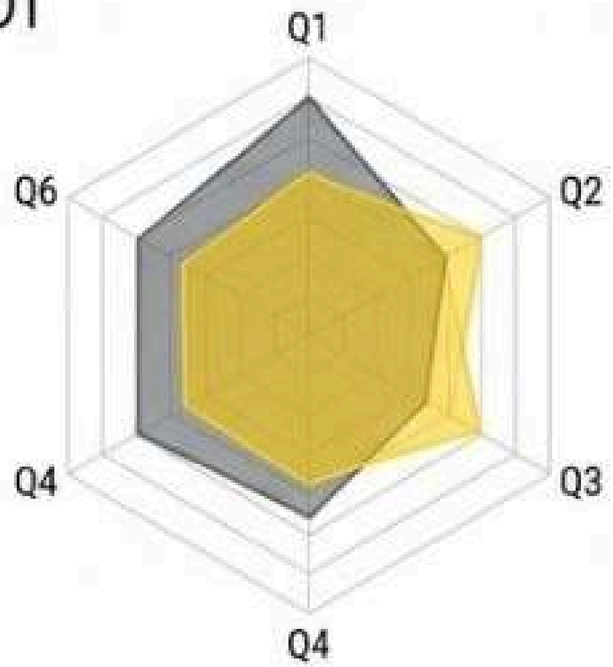


Pain Points addressed

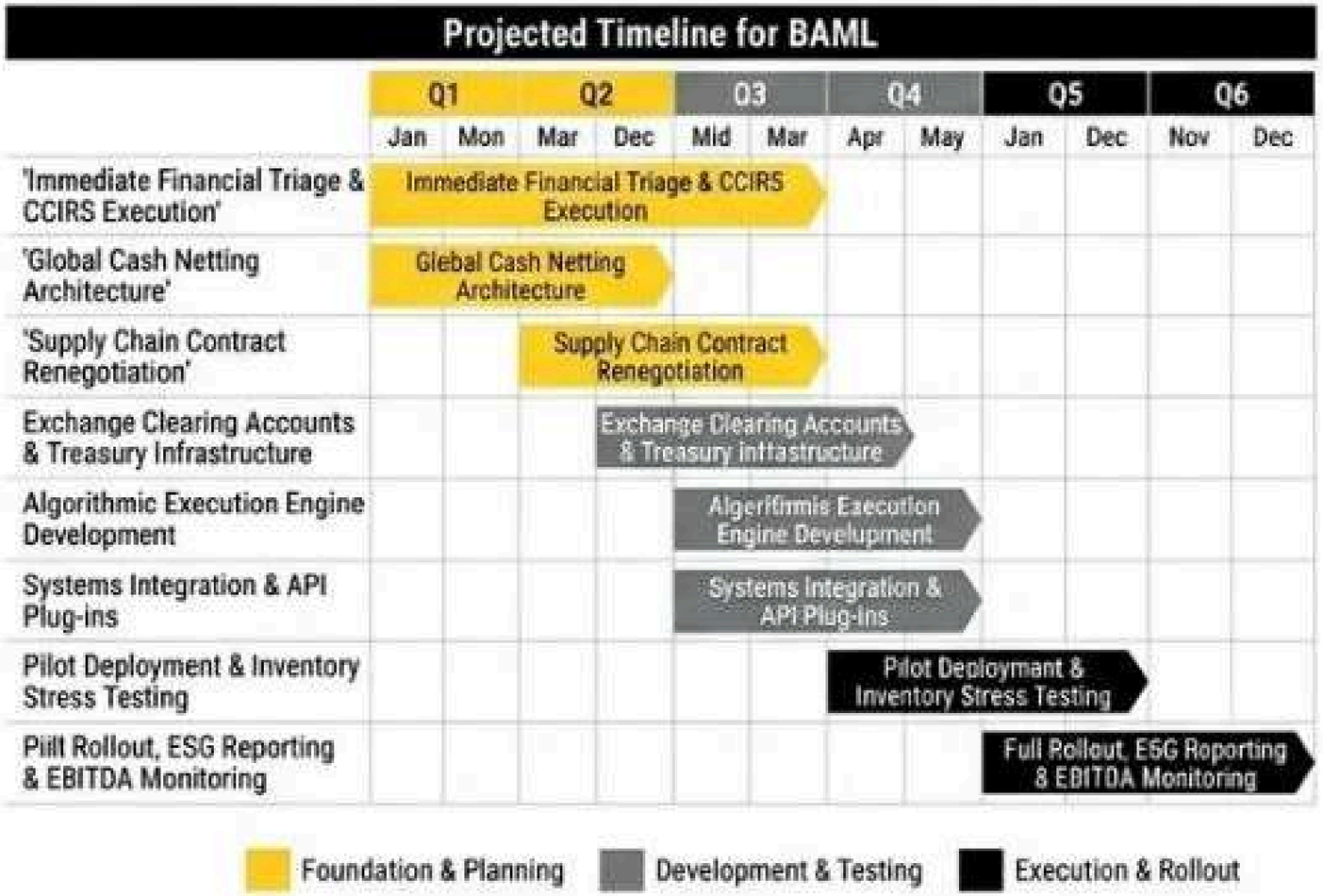
Working Capital Drag	Margin Volatility
FX Translation Loss	ESG Compliance
Interest Rate Spikes	Supply Chain Delays

IMPLEMENTATION ROAD MAP

Q1



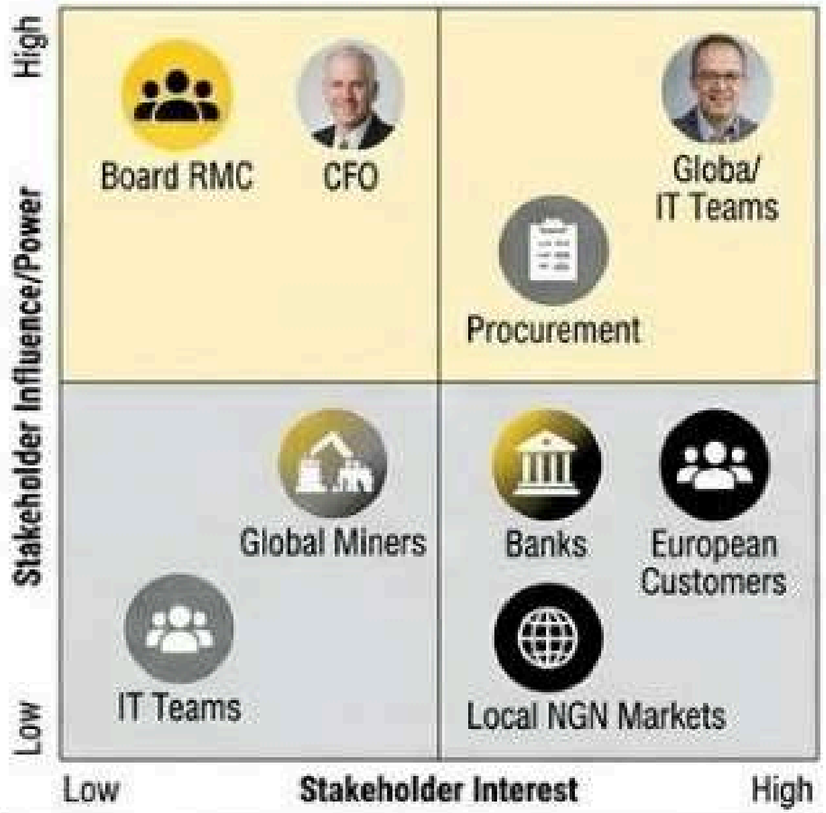
Quarters 1, 2, and 3 are the most significant for establishing financial governance, whereas Quarters 2 and 3 will require maximum attention in resource allocation to integrate treasury APIs and renegotiate global supply chain contracts.



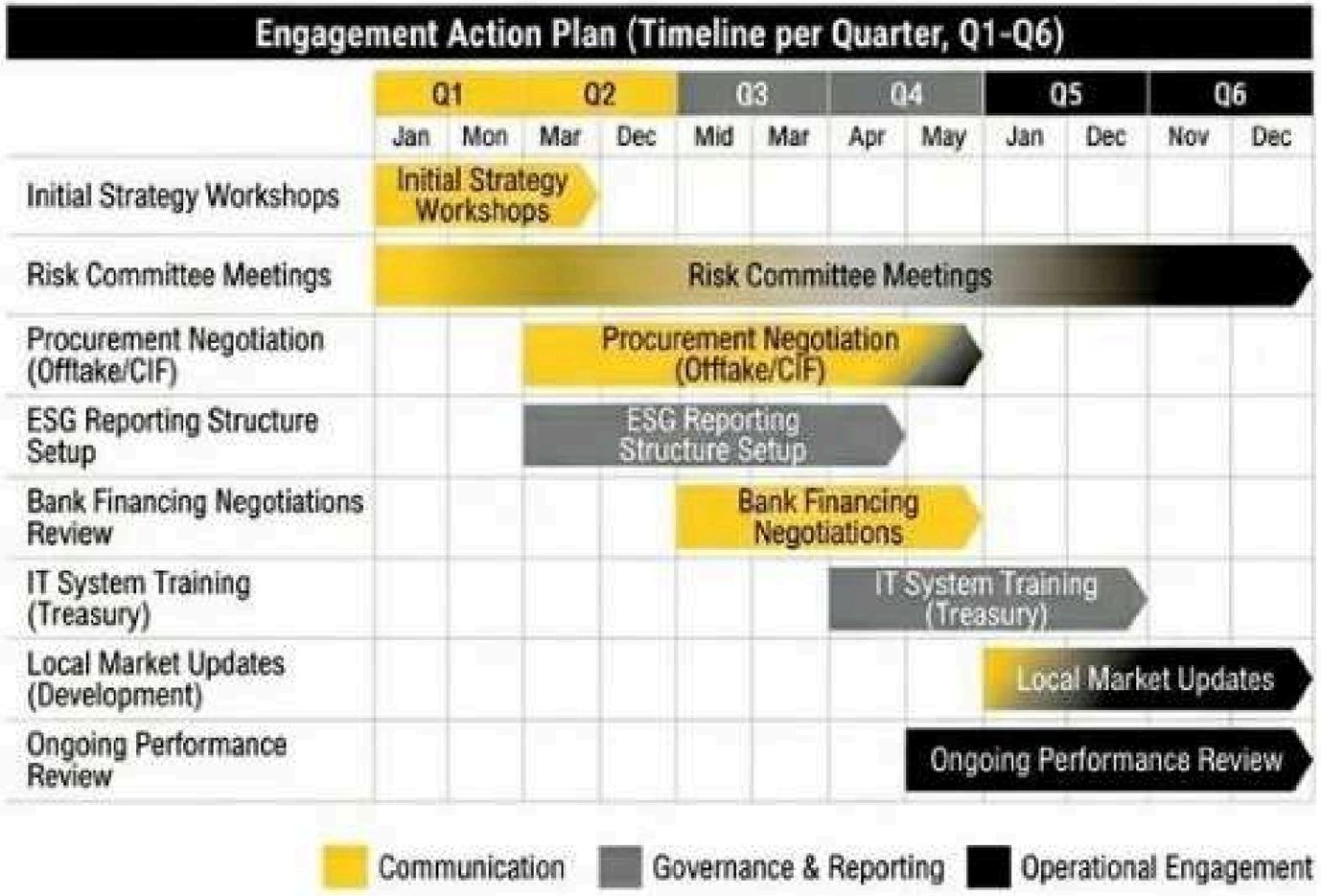
Stakeholder Engagement

STAKEHOLDER ENGAGEMENT & MANAGEMENT FRAMEWORK

01



Establish a robust communication and governance structure to ensure alignment with BAML's strategic goals.

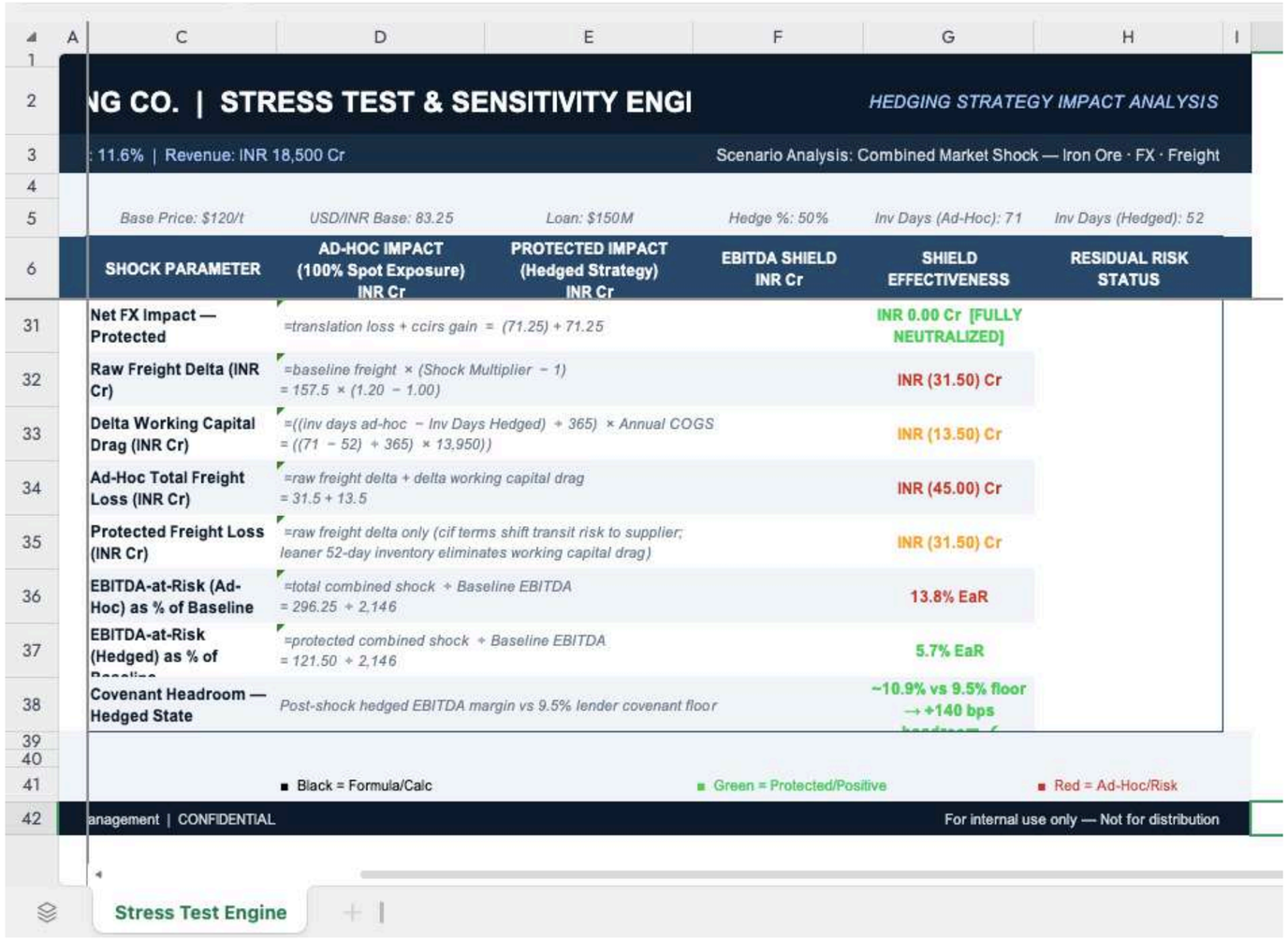


Appendix


CORTEX TREASURY
LIVE
STEELWORKS MFG. CO · PROCUREMENT & RISK DESK
USD/INR 83.42
SOFR 5.31%
LME FE62 $114.80
18:14:51 UTC
MODULE 01
ALGORITHMIC COMMODITY EXECUTION
ALMGREN-CHRISS · TWAP ENGINE · IRON ORE
EXECUTION TRIGGER STATUS
SPOT 7.4% BELOW EMA → MONITORING — THRESHOLD: -8.00%
SIGNAL
STANDBY
IRON ORE SPOT VS 12M EMA · USD/MT · LME
SPOT
12M EMA
SPREAD
$125.52
$116.83
7.44%
ALGO PARAMS
H=0.0025 · Γ=0.001 · T=252
$175
$150
$125
$100
$75
8% TRIGGER
TWAP EXECUTION SLICER · BLOCK ORDER DECOMPOSITION
BLOCK SIZE
ALGO
FILLED
AVG EXEC
50,000 MT
TWAP/10D
4/10 SLICES
$114.32/MT
ACTIVE ORDER
5.3K
5.0K
5.1K
5.0K
5.0K
5.1K
4.6K
4.9K
4.7K
4.7K
D+1
D+2
D+3
D+4
D+5
D+6
D+7
D+8
D+9
D+10
EXECUTED
PENDING
MODULE 02
DYNAMIC INVENTORY PRICING
AVELLANEDA-STOIKOV MODEL · RESERVATION PRICE ENGINE
PHYSICAL INVENTORY LEVEL · DAYS OF COVER
52 DAYS
TARGET: 60D · DEFICIT: 8D
NEUTRAL BUY MODE
40D
TARGET 60D
PASSIVE 70D+
52D
0
15
30
45
60
75
90
RESERVATION PRICE · A-S MODEL OUTPUT
POSTURE
NEUTRAL
INVENTORY RISK SCORE
+13.3%
EXECUTION INSTRUCTION
Bid at mid — MID ± 0.10σ
γ (RISK AVERSION)
0.0012
Σ (PRICE VOL)
2.34%
κ (ORDER INTENSITY)
0.87
INVENTORY TREND · 30-DAY HISTORY
MODULE 03
FX NET SPREAD & DEBT HEDGING
USD FLOW NETTING · CCIRS OVERLAY · SOFR NEUTRALIZATION
FX NET POSITION · USD ANNUAL FLOWS ($M)
USD EXPORTS
$620M
NET SURPLUS
+$140M
USD IMPORTS
$480M
NATURAL HEDGE
COVER RATIO
1.29x
$620M EXPORTS
$480M IMPORTS
DEBT HEDGING · CCIRS OVERLAY
LOAN NOTIONAL
$150M USD
INSTRUMENT
CCIRS 5Y
FLOATING LEG
SOFR + 185bps
FIXED LEG
4.72% INR
FLOATING SOFR RISK NEUTRALIZED
CCIRS CONVERTS USD SOFR FLOAT → INR FIXED · DURATION MATCH 5Y · MTM: +$2.3M
PAY
SOFR+185bps
RECEIVE
4.72% FIXED



https://claude.ai/public/artifacts/db9c7af7-fc66-41fd-a009-b225fb37e117



NG CO. | STRESS TEST & SENSITIVITY ENGI — HEDGING STRATEGY IMPACT ANALYSIS

: 11.6% | Revenue: INR 18,500 Cr — Scenario Analysis: Combined Market Shock — Iron Ore · FX · Freight

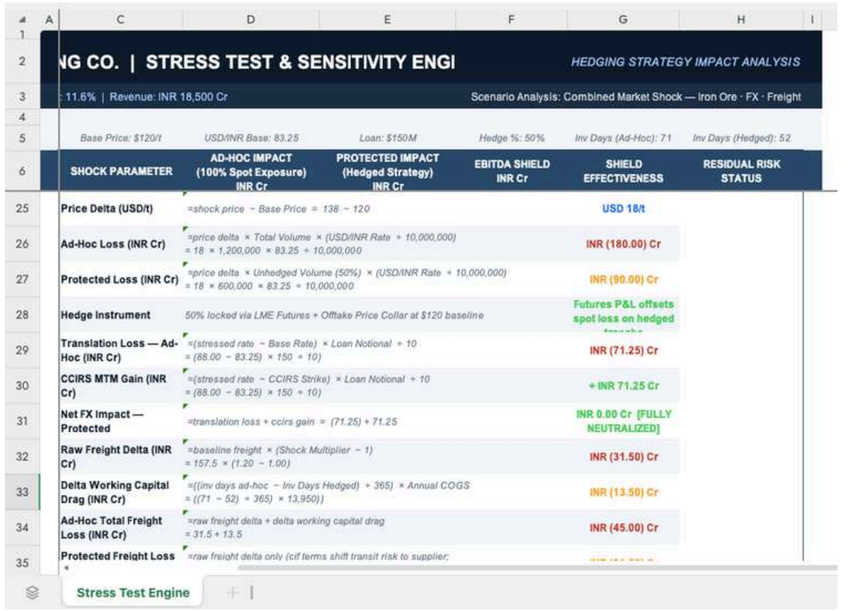
Base Price: $120/t | USD/INR Base: 83.25 | Loan: $150M | Hedge %: 50% | Inv Days (Ad-Hoc): 71 | Inv Days (Hedged): 52

SHOCK PARAMETER	AD-HOC IMPACT (100% Spot Exposure) INR Cr	PROTECTED IMPACT (Hedged Strategy) INR Cr	EBITDA SHIELD INR Cr	SHIELD EFFECTIVENESS	RESIDUAL RISK STATUS
Net FX Impact — Protected	=translation loss + ccirs gain = (71.25) + 71.25			INR 0.00 Cr [FULLY NEUTRALIZED]	
Raw Freight Delta (INR Cr)	=baseline freight × (Shock Multiplier − 1) = 157.5 × (1.20 − 1.00)			INR (31.50) Cr	
Delta Working Capital Drag (INR Cr)	=((inv days ad-hoc − Inv Days Hedged) ÷ 365) × Annual COGS = ((71 − 52) ÷ 365) × 13,950))			INR (13.50) Cr	
Ad-Hoc Total Freight Loss (INR Cr)	=raw freight delta + delta working capital drag = 31.5 + 13.5			INR (45.00) Cr	
Protected Freight Loss (INR Cr)	=raw freight delta only (cif terms shift transit risk to supplier; leaner 52-day inventory eliminates working capital drag)			INR (31.50) Cr	
EBITDA-at-Risk (Ad-Hoc) as % of Baseline	=total combined shock ÷ Baseline EBITDA = 296.25 ÷ 2,146			13.8% EaR	
EBITDA-at-Risk (Hedged) as % of Baseline	=protected combined shock ÷ Baseline EBITDA = 121.50 ÷ 2,146			5.7% EaR	
Covenant Headroom — Hedged State	Post-shock hedged EBITDA margin vs 9.5% lender covenant floor			~10.9% vs 9.5% floor → +140 bps	

■ Black = Formula/Calc ■ Green = Protected/Positive ■ Red = Ad-Hoc/Risk

anagement | CONFIDENTIAL — For internal use only — Not for distribution

Stress Test Engine



NG CO. | STRESS TEST & SENSITIVITY ENGI — HEDGING STRATEGY IMPACT ANALYSIS

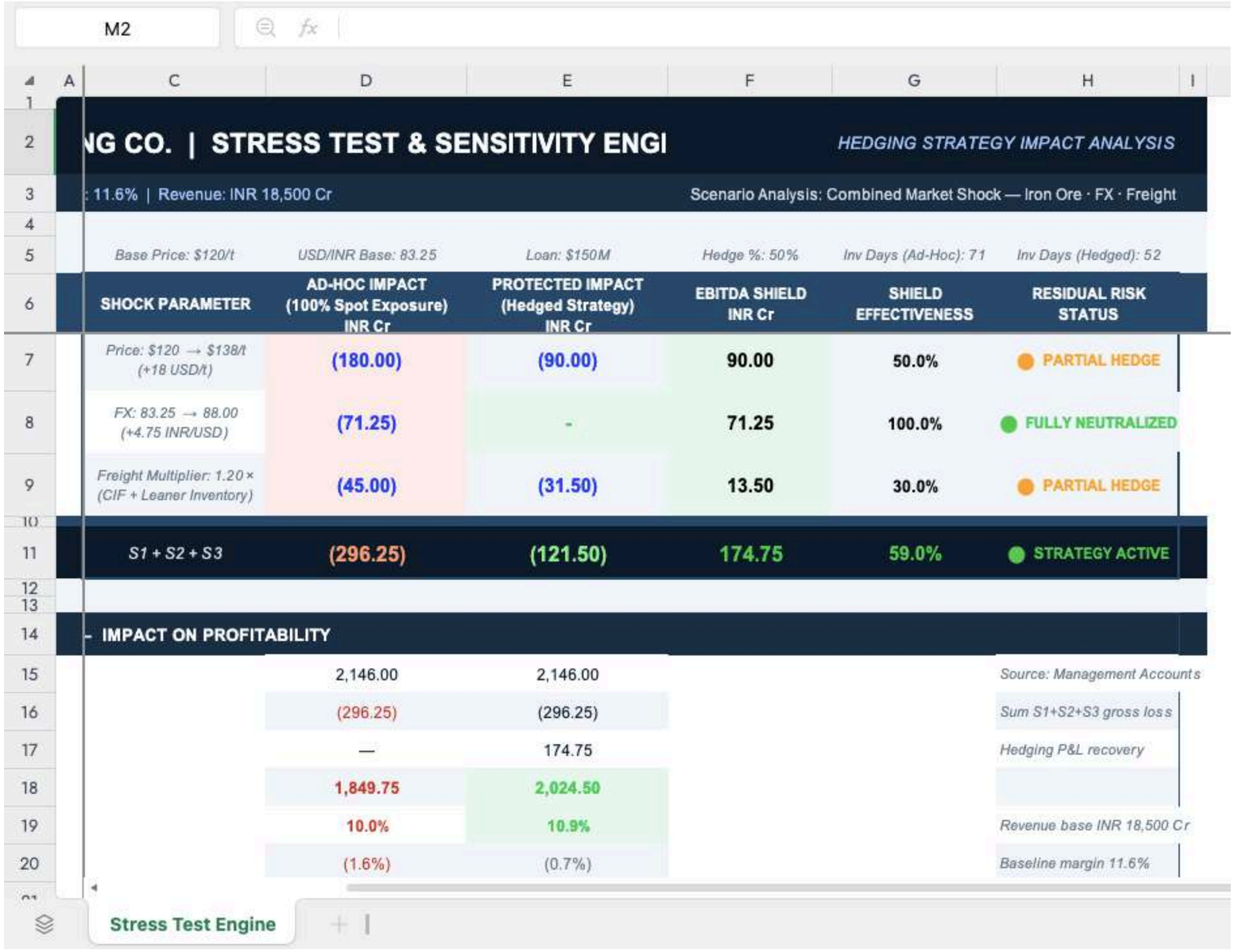
: 11.6% | Revenue: INR 18,500 Cr — Scenario Analysis: Combined Market Shock — Iron Ore · FX · Freight

Base Price: $120/t | USD/INR Base: 83.25 | Loan: $150M | Hedge %: 50% | Inv Days (Ad-Hoc): 71 | Inv Days (Hedged): 52

SHOCK PARAMETER	AD-HOC IMPACT (100% Spot Exposure) INR Cr	PROTECTED IMPACT (Hedged Strategy) INR Cr	EBITDA SHIELD INR Cr	SHIELD EFFECTIVENESS	RESIDUAL RISK STATUS
Price Delta (USD/t)	=shock price − Base Price = 138 − 120			USD 18/t	
Ad-Hoc Loss (INR Cr)	=price delta × Total Volume × (USD/INR Rate ÷ 10,000,000) = 18 × 1,200,000 × 83.25 ÷ 10,000,000			INR (180.00) Cr	
Protected Loss (INR Cr)	=price delta × Unhedged Volume (50%) × (USD/INR Rate ÷ 10,000,000) = 18 × 600,000 × 83.25 ÷ 10,000,000			INR (90.00) Cr	
Hedge Instrument	50% locked via LME Futures + Offtake Price Collar at $120 baseline			Futures P&L offsets spot loss on hedged	
Translation Loss — Ad-Hoc (INR Cr)	=(stressed rate − Base Rate) × Loan Notional ÷ 10 = (88.00 − 83.25) × 150 ÷ 10)			INR (71.25) Cr	
CCIRS MTM Gain (INR Cr)	=(stressed rate − CCIRS Strike) × Loan Notional ÷ 10 = (88.00 − 83.25) × 150 ÷ 10)			+ INR 71.25 Cr	
Net FX Impact — Protected	=translation loss + ccirs gain = (71.25) + 71.25			INR 0.00 Cr [FULLY NEUTRALIZED]	
Raw Freight Delta (INR Cr)	=baseline freight × (Shock Multiplier − 1) = 157.5 × (1.20 − 1.00)			INR (31.50) Cr	
Delta Working Capital Drag (INR Cr)	=((inv days ad-hoc − Inv Days Hedged) ÷ 365) × Annual COGS = ((71 − 52) ÷ 365) × 13,950))			INR (13.50) Cr	
Ad-Hoc Total Freight Loss (INR Cr)	=raw freight delta + delta working capital drag = 31.5 + 13.5			INR (45.00) Cr	
Protected Freight Loss	=raw freight delta only (cif terms shift transit risk to supplier;				

Stress Test Engine

https://in.docworkspace.com/d/sbCaehM5zveUupsx_l2rk5abs9uzwxu3w56



M2

NG CO. | STRESS TEST & SENSITIVITY ENGI — HEDGING STRATEGY IMPACT ANALYSIS

: 11.6% | Revenue: INR 18,500 Cr — Scenario Analysis: Combined Market Shock — Iron Ore · FX · Freight

Base Price: $120/t	USD/INR Base: 83.25	Loan: $150M	Hedge %: 50%	Inv Days (Ad-Hoc): 71	Inv Days (Hedged): 52

SHOCK PARAMETER	AD-HOC IMPACT (100% Spot Exposure) INR Cr	PROTECTED IMPACT (Hedged Strategy) INR Cr	EBITDA SHIELD INR Cr	SHIELD EFFECTIVENESS	RESIDUAL RISK STATUS
Price: $120 → $138/t (+18 USD/t)	(180.00)	(90.00)	90.00	50.0%	PARTIAL HEDGE
FX: 83.25 → 88.00 (+4.75 INR/USD)	(71.25)	-	71.25	100.0%	FULLY NEUTRALIZED
Freight Multiplier: 1.20× (CIF + Leaner Inventory)	(45.00)	(31.50)	13.50	30.0%	PARTIAL HEDGE
S1 + S2 + S3	(296.25)	(121.50)	174.75	59.0%	STRATEGY ACTIVE

IMPACT ON PROFITABILITY

2,146.00	2,146.00		Source: Management Accounts
(296.25)	(296.25)		Sum S1+S2+S3 gross loss
—	174.75		Hedging P&L recovery
1,849.75	2,024.50		
10.0%	10.9%		Revenue base INR 18,500 Cr
(1.6%)	(0.7%)		Baseline margin 11.6%

Stress Test Engine

Thank You